

Mihal Dimo

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EDUCATION

University of Catania

September 2021 – Present (Expected June 2027)

Bsc in Computer Science

Relevant Modules: **Embedded systems, Algorithm & Data Structures, Linear Algebra**

Brandenburgische Technische Universität Cottbus

2024 – 2024

Bsc in Computer Science

Exchange Study Erasmus+

Relevant Modules: **Applied Linear Algebra for AI, Calculus, Software Security**

WORK EXPERIENCE

Open Source Contributor

July 2026 – Present

vLLM Project

- Contributed to **20+ merged PRs** in the vLLM Metal backend, spanning **speculative decoding, paged KV-cache management, model compatibility, sampling, and memory management** for Apple Silicon inference.
- Implemented **DSpark speculative decoding** for the vLLM Metal backend, including its Qwen3 model integration, scheduler-owned KV-cache management, paged execution, Markov drafting, prefix caching, validation, and serving benchmarks.
- Optimized **speculative decoding** by eliminating redundant KV re-ingestion and chunking cold draft-KV ingestion, reducing **8K first-propose latency** $31\times$ (**3.47s** $\rightarrow$ **112ms**), E2E latency **2.1 $\times$** (**64.0** $\rightarrow$ **30.8 ms/token**), steady-state KV ingestion by **50%**, and DSpark draft-weight memory by **48%** (**4.42** $\rightarrow$ **2.29 GiB**).

Software Development Engineer Intern

January 2026 – July 2026

Amazon

Berlin, Germany

- Engineered a **full-stack embedding search and visualization platform** over **50M embeddings**, with PCA/UMAP visualization and sub-**5 ms KNN queries**, accelerating Music recommender experimentation.
- Designed a **batch ML data pipeline** for embedding metadata hydration and model indexing across **S3, ECS, and OpenSearch**, enabling automated weekly model updates.
- Built **evaluation workflows** measuring marketplace performance with **Recall, NDCG, Precision, and triplet accuracy**, alongside a **FastAPI backend** and CloudWatch monitoring.

Firmware Lead & Embedded Software Engineer

March 2024 – December 2025

KBDFans (Partner)

Changzhou, Jiangsu, China

- Lead firmware and hardware development for the **Zellia Hall Effect project** (team of 5 C/C++ engineers), designing a **distributed embedded system** across 5 AT32 MCUs with enhanced modularity and multi-layout keyboard using **KiCad 8.0**.
- Achieved **<0.28 ms input latency** with **106 kHz Scan Rate** by decoupling signal acquisition and processing, and engineered a custom **7.5 Mbps UART protocol** for high-throughput multi-MCU synchronization.
- Offloaded **ADC normalization** and calibration to slave MCUs, reducing master CPU load by **90%** and enabling faster processing.

PROJECTS

Custom FP8 CUDA Kernel – Qwen3.8-27B on H100

github.com/mhdimo/qwen38-h100-lab

- Wrote an **SM90 CUDA kernel** fusing SiLU, gating, and per-group **FP8 E4M3 quantization**, using warp-shuffle-only reductions with no block barriers; verified **byte-exact** against vLLM's kernel across 168 reference comparisons (edge tiles, both scale layouts) and validated **CUDA graph** replay with changing inputs.
- Profiled the serving stack to target the kernel (14.5% of prefill GPU time), reaching **2.26 $\times$** kernel speedup at the 16K-row shape and **+5.90% prefill / +3.65% mixed** full-model throughput vs. tuned vLLM.

Inference Engine — C++20 Metal LLM Runtime

github.com/mhdimo/inference-engine

- Built a **from-scratch C++20 LLM inference engine** for Apple Silicon, implementing tensor runtime, compute graphs with **lifetime-based memory planning**, GGUF loading, tokenization, KV caching, sampling, and CPU/Metal backends without ML frameworks.
- Implemented **13 custom Metal compute kernels** with zero-copy unified-memory execution, including tiled and quantized GEMM, RoPE, GQA attention, RMSNorm, and SwiGLU, enabling end-to-end inference of **Qwen2.5 and SmolLM2**.

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, TypeScript, Java, SQL.

ML Systems: vLLM, MLX (custom Metal kernels), PyTorch, Metal/MPS backends, speculative decoding, quantization, KV-cache optimization, embeddings, recommender systems, DQN.

LLM Tooling: MCP, multi-provider LLM APIs, AI-SDK, Claude code.

Systems & Data: Linux, Docker, AWS, REST APIs, Kafka, PySpark, OpenSearch, Parquet, Pandas, Numpy.

Embedded & Web: ARM Cortex-M, KiCad, USB/WebUSB, FastAPI, Node.js, SvelteKit, React.